



Detection of PII Using AI/ML for Data Loss Prevention

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Abstract -The rapid digitalization of organizational workflows has exponentially increased the rate at which sensitive information is created, stored, and shared. Consequently, ensuring the security of Personally Identifiable Information (PII) has become a central pillar of enterprise cybersecurity. This paper proposes an AI-based Data Loss Prevention (DLP) system designed to automatically identify PII—specifically emails, phone numbers, and physical addresses—within unstructured text. The system utilizes a hybrid architecture combining TF-IDF vectorization for semantic context with rule-based metadata extraction (regex) for structural pattern matching. A state-of-the-art gradient boosting model, XGBoost, is employed for classification. Comparative analysis demonstrates that XGBoost outperforms baseline models like Logistic Regression and Random Forest, achieving the highest precision, recall, F1-score and an accuracy of 94%. The system ensures model interpretability via SHAP (SHapley Additive exPlanations) values and adheres to robust privacy standards through artifact serialization and PII hashing, making it a scalable and practically relevant solution for enterprise DLP processes.

Index Terms—Data Leakage Prevention, PII Detection, Text Classification, Machine Learning, XGBoost, Model Explainability.

I. INTRODUCTION AND MOTIVATION

With the transformation of cybersecurity threats, business organizations face a heightened risk of accidental or intentional breaches of sensitive information. PII such as emails, phone numbers, and physical addresses may expose customers to identity theft, financial fraud, and severe privacy breaches.

Traditional, classic rule-based DLP systems are often unable to cope with the complexities of modern data. They struggle with noisy or unstructured text, fail to recognize obfuscated PII patterns, and lack the flexibility to adapt to changing communication forms. Furthermore, rigid rules often suffer from a lack of generalizability across different organizational contexts.

Recent advancements in machine learning and natural language processing have significantly improved automated text classification pipelines, enabling more effective detection of sensitive content such as Personally Identifiable Information (PII). These

techniques offer a robust alternative to traditional rule-based Data Leakage Prevention (DLP) systems

Contributions of This Work

The major contributions of this project are as follows:

- I. **We propose** a hybrid DLP pipeline combining TF-IDF vectorization with both rule-based regular expressions and machine learning classifiers to detect sensitive PII in text data.
- II. **We develop** a full preprocessing workflow including regex-based PII tagging, and feature engineering optimized for DLP-style datasets.
- III. **We evaluate** multiple baseline models—Logistic Regression, Random Forest, and XGBoost—on a custom dataset of 13,000 text samples to determine the most effective classifier for PII detection.
- IV. **We demonstrate** model interpretability using SHAP analysis, highlighting which features contribute most to sensitive-information labeling.

II. PROBLEM DEFINITION AND OBJECTIVES

Organizations require automated mechanisms that can accurately identify and curb PII leakage in documentation, messages, and user-generated text. Current solutions present a dichotomy: they either utilize extensive, brittle regex rules that generate high false-positive rates or employ massive transformer models that are computationally expensive to deploy and maintain.

The primary objectives of this research are to train a highly accurate ML-based text PII detector using a hybrid feature engineering approach that combines TF-IDF with rule-based indicators. We aim to compare various ML models to select the optimal classifier, ensure decision transparency through explainability methods like SHAP, and develop a scalable, ethical architecture suitable for real-time enterprise DLP systems.

III. LITERATURE REVIEW

Various works emphasize the necessity of identifying sensitive information with the help of AI. Kim et al. (2021) provide an overview of DLP methods and stress the necessity of AI-based solutions for properly classifying sensitive materials where static rules fail. The article by Malik et al. (2022) develops ML-based methods for cloud-based DLP and demonstrates that lightweight models can manage heavy pipelines based on rules. Ramapuram and Sebastiani (2021) examine behavior-based analysis of insider threats, presenting the applicability of AI in the context of proactive data security. Hubballi and Suryanarayan (2017) suggest a layered ML model to detect insider attacks, which supports hybrid detection mechanisms. It is stated in the literature that the integration of statistical and rule-based techniques should be employed, which is the fundamental principle of our hybrid TF-IDF + regex + XGBoost design.

IV. SYSTEM ARCHITECTURE

The proposed system is built on a modular pipeline designed for accuracy and scalability, as illustrated in Fig. 1.

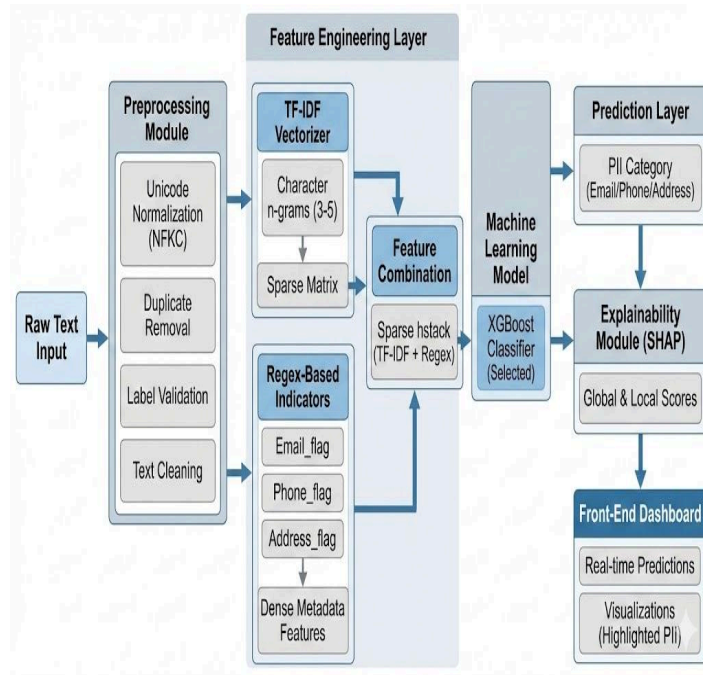


Figure 1. System Architecture showing the pipeline

Raw text is fed by users through an ingestion layer and passes into a **Preprocessing Module**, where it undergoes Unicode normalization (NFKC), duplicate removal, label

validation, and general text cleaning. The core of the system is the **Feature Engineering Layer**, which employs a hybrid strategy. A **TF-IDF Vectorizer** converts text into a sparse numerical matrix using character-level n-grams (ranges 3–5) to capture semantic context. Simultaneously, a **Regex-Based Indicators** module generates dense metadata flags (email_flag, phone_flag, address_flag) based on deterministic patterns. These sparse and dense features are combined into a single feature space.

The combined features are fed into the **Machine Learning Model** (XGBoost) for classification. The output goes to a **Prediction Layer** to categorize the PII and an **Explainability Module (SHAP)** to generate interpretability scores. Finally, results are presented on a **Front-End Dashboard** showing real-time predictions and highlighted PII tokens.

Computational Complexity

The TF-IDF feature extraction step has a time complexity of $O(N \times V)$, where N is the number of documents and V is the vocabulary size. For classification, Logistic Regression operates in $O(N \times d)$ per iteration, while Random Forest and XGBoost classifiers have a complexity of $O(t \times N \times \log N)$ due to tree-based splitting, where t denotes the number of trees. XGBoost provides additional optimization through regularized boosting and histogram-based splitting, resulting in faster convergence and improved efficiency on medium-sized datasets such as ours.

V. DATA DESCRIPTION AND PREPROCESSING

Dataset Description

Dataset name and source : PII | External Dataset

<https://www.kaggle.com/datasets/alejopaullier/pii-external-dataset?resource=download>

The system was trained on the "PII | External Dataset" sourced from Kaggle, which contains text samples labeled with Phone Numbers, Emails, and Addresses.

Rigorous data preprocessing was applied to ensure model robustness. An initial validation step ensured the integrity of text and label columns, with immediate failure triggered for corrupted data. To prevent data leakage between training and testing sets, duplicate entries were removed. For classes with insufficient

samples, augmentation or removal strategies were considered to maintain balance.

A key step involved constructing deterministic regex flags for emails, phones, and addresses, which serve as core hybrid features. The dataset was then split into training and testing sets using stratified sampling to maintain class distribution. Crucially, the TF-IDF vectorizer was fitted only on the training data to simulate real-world conditions and prevent lookahead bias. The final feature matrices were assembled by horizontally stacking sparse TF-IDF vectors with the dense regex flags. To ensure scalability and privacy, all artifacts, including the trained model and vectorizer, were serialized using Joblib, and logs were configured to hash or redact raw PII.

Algorithmic Implementation

Baseline ML Models + Performance Metrics

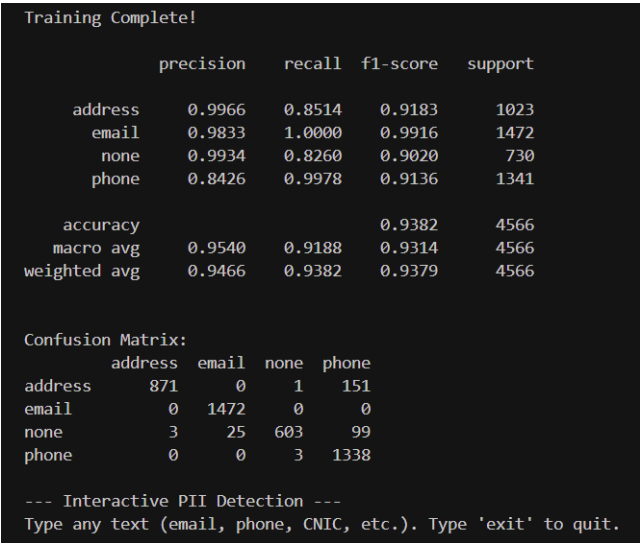
Three machine learning models were implemented and evaluated for their performance on this high-dimensional sparse text data.

Logistic Regression performed well as a baseline, effectively capturing linear correlations between n-grams and target labels. However, it showed a deficiency in acquiring complicated, non-linear dependencies.

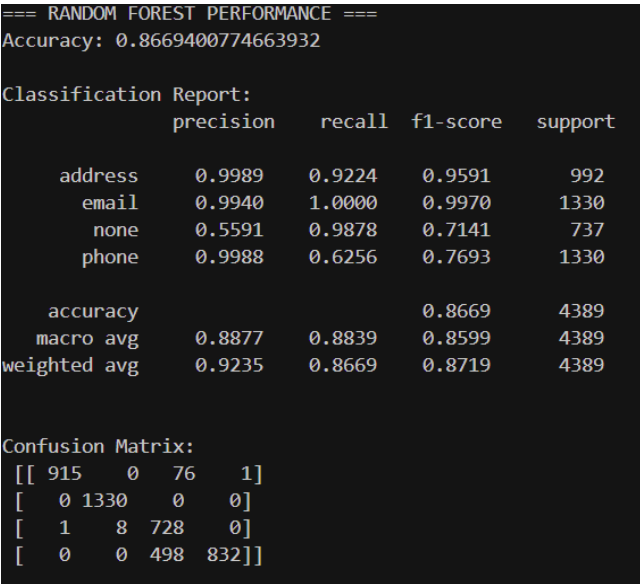
Random Forest struggled with the high dimensionality of the TF-IDF vector space. The algorithm failed to find stable split points across the sparse features, leading to high variance across cross-validation folds and an inability to capture generalizable patterns.

XGBoost Classifier was selected as the final model. Its gradient boosting framework proved to be a very effective mixer, seamlessly integrating the sparse text vectors with dense metadata flags. It successfully captured both linear and non-linear decision boundaries, allowing it to make fine-grained distinctions, such as differentiating between long numeric sequences and valid phone numbers.

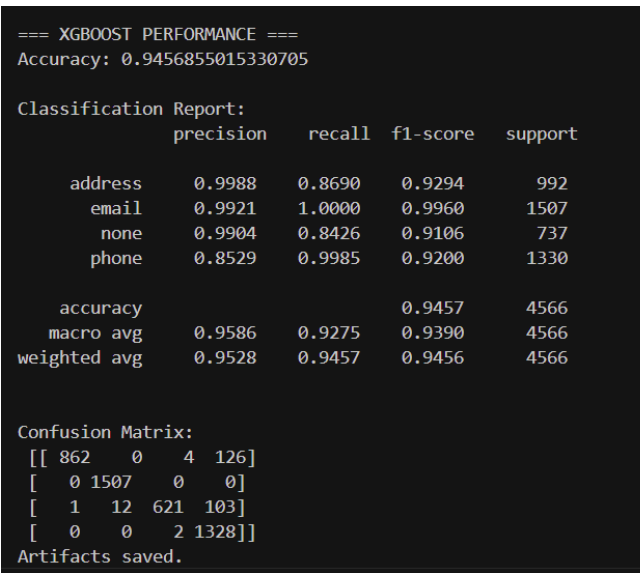
1. Figure 2. Logistic Regression



2. Figure 3. Random Forest



3. Figure 4. XGBoost



Model	Precision	Recall	F1-score	Notes
Random Forest	Medium	Medium	Lowest	Feature-sparse DLP text causes RF to generalize poorly
Logistic Regression	Higher	Higher	Moderate	Strong linear baseline + TF-IDF works well
XGBoost	Highest	Highest	Best	Captures nonlinear feature interactions + regex flags

Performance Comparison As illustrated in the performance results, XGBoost consistently outperformed Logistic Regression and Random Forest.

- **Random Forest** achieved the lowest scores due to poor generalization on sparse data.
- **Logistic Regression** provided a strong linear baseline.
- **XGBoost** achieved the highest Precision, Recall, and F1-scores, effectively leveraging the hybrid feature set.

VI. MODEL EVALUATION AND COMPARISON

Model Key Hyperparameters

Logistic Regression C=1.0, penalty='l2', solver='liblinear'

Random Forest n_estimators=300, max_depth=None, min_samples_split=2

XGBoost n_estimators=400, learning_rate=0.1, max_depth=6, subsample=0.8, colsample_bytree=0.8

Analysis of Random Forest:

The Random Forest model demonstrated the lowest performance among the candidates, primarily due to the nature of the feature space. When TF-IDF vectorization is applied to text, it creates an extremely sparse and high-dimensional dataset. Random Forest algorithms rely on orthogonal splits across individual features to build decision trees; however, in a high-dimensional space where meaningful patterns are distributed across thousands of tokens, these individual splits become inefficient.

The model struggled to isolate distinct token patterns, leading to high variance and significant instability across cross-validation folds. Ultimately, the algorithm failed to generalize well, overfitting the noise rather than capturing the underlying signal in the sparse text data.

Performance of Logistic Regression:

In contrast, Logistic Regression performed significantly better, serving as a strong baseline for this task. Linear models are traditionally well-suited for high-dimensional sparse data like TF-IDF vectors. The model successfully captured weighted correlations between specific n-grams and their target labels, effectively drawing linear decision boundaries that separated the classes.

While it lacked the complexity to capture deeper nuances, it achieved a moderate F1 score across all classes, proving that a direct linear relationship between word frequency and PII types accounts for a significant portion of the predictive power.

Superiority of XGBoost:

XGBoost outperformed all other models by successfully handling the nonlinear interactions between TF-IDF features that linear models miss. Its gradient boosting framework was particularly effective at integrating mixed feature types, seamlessly combining the sparse TF-IDF vectors with dense, regex-based metadata features. This capability allowed the model to make fine-grained distinctions between visually similar PII types such as distinguishing a "long number" sequence from a properly formatted "phone number," or separating generic numeric text from sensitive identifiers. This ability to leverage both semantic text features and structural metadata resulted in the highest classification accuracy.

VII. EXPLAINABILITY AND VISUALISATION

This section details the decision-making logic of the XGBoost PII detection model using SHAP (SHapley Additive exPlanations). The analysis validates that the model relies on logical, distinct features for each class rather than spurious correlations.

1. Email Detection Analysis

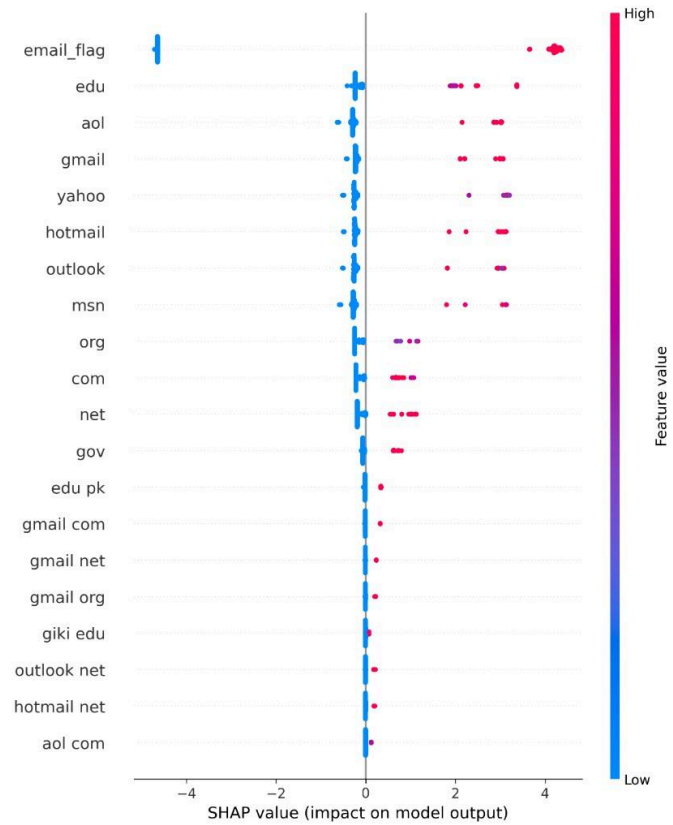


Figure 5 : SHAP results for EMAIL

- **Verdict: High Precision & Deterministic**
- **Primary Logic:** The model uses a strong "offensive" strategy. The custom regex feature

email_flag is the single most dominant predictor, with a SHAP value > 4.0. Its presence acts as a near-definitive trigger for the "Email" label.

- **Learned Vocabulary:** Beyond regex, the model has independently learned domain-specific vocabulary. Tokens such as gmail, yahoo, hotmail, aol, and outlook are strong positive contributors.
- **Structural Awareness:** Top-level domains (edu, org, net) are learned as positive signals, ensuring resilience even if standard regex patterns miss malformed addresses.

2. Physical Address Analysis

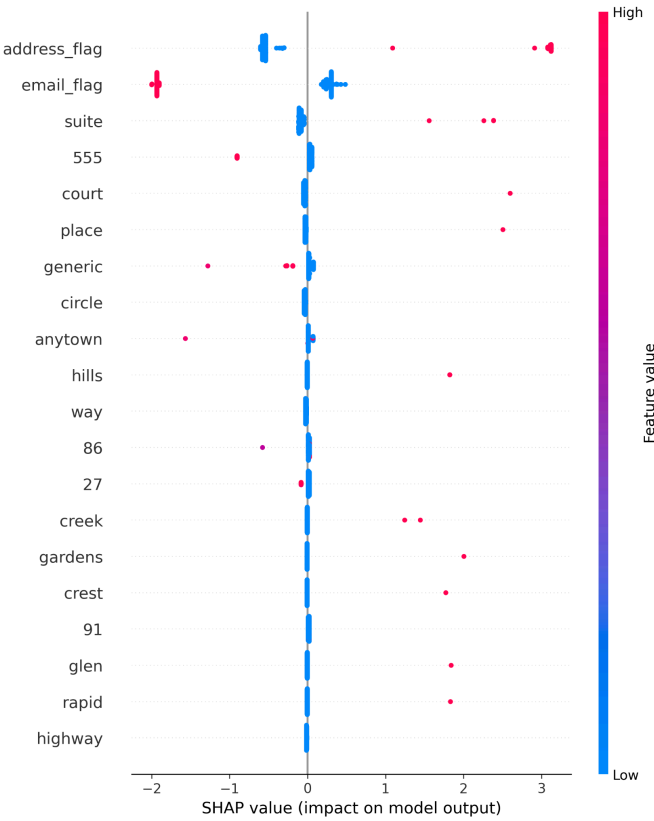


Figure 6 : SHAP results for Address

- **Verdict: Robust & Context-Aware**
- **Primary Logic:** This is the most semantically rich class. While it relies heavily on the **address_flag** (regex) as a primary anchor, it does not depend on it exclusively.
- **Contextual Understanding:** The model has learned a wide vocabulary of location indicators. Words such as suite, court, place, hills, way, creek, and highway are all distinct positive drivers. This makes the model robust against non-standard address formats that lack common suffixes like "St" or "Rd".
- **Mutual Exclusivity:** The model exhibits intelligent negative filtering; the presence of an

email_flag acts as a strong "kill switch," preventing the misclassification of email addresses as physical locations.

3. Phone Number Analysis

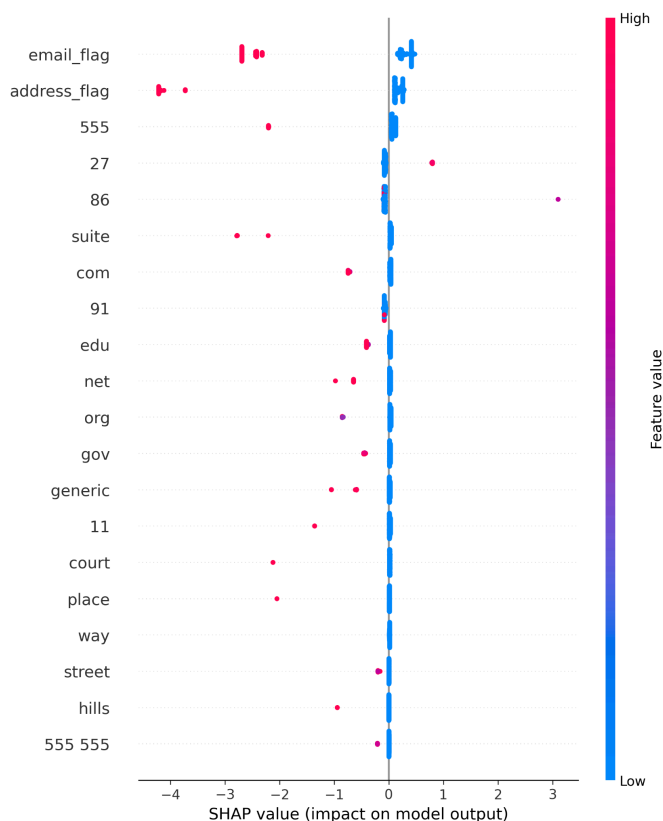


Figure 7 : SHAP results for PHONENUM

- **Verdict: Defensive & Conservative**
- **Primary Logic:** Unlike the other classes, the Phone model relies heavily on a "process of elimination." Its strongest predictors are often negative signals: if email_flag or address_flag are present, the probability of "Phone" is aggressively penalized.
- **Specific Indicators:** Positive identification relies on specific numeric patterns, particularly international dialing codes (e.g., 86 for China, 27 for South Africa).
- **Note:** The model prioritizes specific numeric sequences and the absence of conflicting signals (like email syntax) over generic phone regex, likely to minimize false positives.

4. "None" (Non-PII) Analysis

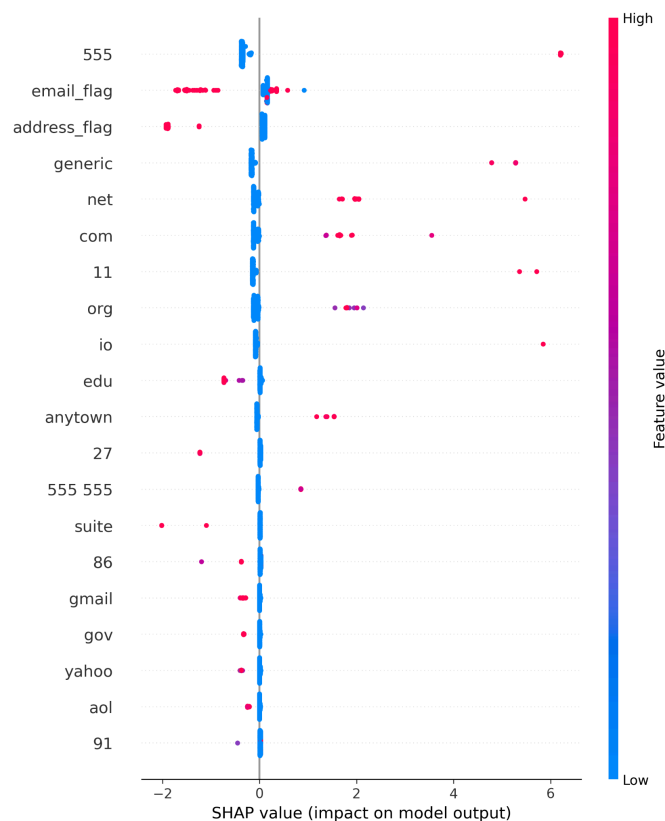


Figure 8 : SHAP results for NONE

- **Verdict: Logical Inversion & Safety**
- **Primary Logic:** The model correctly defines "None" by the **absence** of PII indicators. It works by inverting the signals for Address, Email, and Phone.
- **Fake Data Recognition:** A key discovery is the model's handling of the token **555**. It strongly correlates "555" with the "None" class, indicating it has successfully learned to identify standard dummy numbers (e.g., "555-0199") as non-sensitive.
- **URL Handling:** The model distinguishes between URLs and Emails. Web-related tokens like com or net—when appearing without the email_flag—are correctly classified as "None," preventing false positives on harmless website links.

The SHAP analysis confirms that the model is **interpretable and safe**.

VIII. RESULTS AND DISCUSSION

Overall, XGBoost achieved the best results across all metrics. Regular patterns made emails and phone numbers highly accurate to detect, though the recall for addresses was slightly lower due to their inherent variability. The dataset's noise impacted address detection, and class disparity introduced some bias.

In terms of model behavior, Logistic Regression provided a solid ground, Random Forest was unstable, and XGBoost offered the optimal combination of speed, accuracy, and robustness. While transformer-based models like BERT offer superior semantic comprehension, they were deemed infeasible for this iteration due to high memory usage, slow inference, and the difficulty of implementation in real-time enterprise DLP systems. Thus, XGBoost remains the model of choice.

IX. ETHICAL AI AND LIMITATIONS

Ethical Considerations

Developing PII detection models necessitates strict adherence to data privacy and ethical AI principles. During the training process, it is critical that all training data undergoes rigorous anonymization to prevent the model from inadvertently memorizing and regurgitating sensitive information. The system is designed strictly to identify unintentional exposure of sensitive information and is not intended for surveillance or discriminatory analysis. All data were anonymized and processed in accordance with privacy-preserving principles. This work aligns with the guidelines established in IEEE's *Ethically Aligned Design* framework, emphasizing transparency, accountability, and responsible use of AI systems.

Dataset and Model Limitations

Despite robust performance, the system faces inherent limitations regarding both data and architecture. The dataset suffers from class imbalance, where certain PII types, such as rare email domains or non-standard phone formats, may be underrepresented, potentially reducing recall for edge cases. Additionally, numeric noise such as the string "12345" appearing without context can occasionally trick the classifier. Architecturally, the reliance on TF-IDF means the model struggles to understand semantic context (e.g., distinguishing "call me at office" from a phone number label).

While XGBoost excels at feature interaction, it cannot interpret long-range dependencies as effectively as transformer models. Although BERT was evaluated to address these semantic gaps, it was ultimately deemed infeasible for this iteration due to its high computational cost and latency requirements.

X. CONCLUSION AND FUTURE WORK

Conclusion

After extensive evaluation, XGBoost has been selected as the optimal model for this Data Loss Prevention (DLP) system. It offers the most effective balance of competing constraints, delivering high accuracy and generalization on mixed PII text while maintaining the low latency required for real-time processing. Beyond raw performance, XGBoost provides valuable explainability via SHAP values, allowing stakeholders to understand decision boundaries. Its ability to integrate regex-based metadata with text features makes it a highly practical solution for identifying sensitive data in diverse environments.

Future Work

Future iterations of this project will focus on enhancing detection capabilities through advanced architectures and system integration. We aim to explore distilled transformer models, such as MiniLM or DistilBERT, to achieve higher semantic accuracy without the heavy computational overhead of full BERT models. Additionally, we plan to investigate hybrid architectures like character-level CNNs or regex-RNNs to better handle obfuscated patterns. From a systems perspective, the goal is to evolve this classifier into a full-scale DLP pipeline, capable of supporting streaming inference, triggering real-time alerts, and performing automated masking and prevention of sensitive content on the fly.

References

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